

Kidney Cancer

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KEY POINTS

- Renal cell carcinoma (RCC) is diagnosed in approximately 81,000 patients per year in the United States.
- Many more patients will be incidentally found to have a renal mass, prompting an evaluation for RCC.
- The incidence of RCC is increasing in the United States and worldwide.
- Clear cell renal cell carcinoma (ccRCC) is the most common RCC histology.
- RCC is commonly diagnosed in the nephrology clinic during workup of acute kidney injury, obstruction, and/or hematuria.
- Treatment of localized kidney cancer is associated with long-term loss of kidney function.
- Combination treatments targeting VEGF receptor (VEGFR) and/or checkpoint inhibitors have improved survival for advanced kidney cancer.

Cancer arising in the kidney is common, representing the sixth most common cancer in men and the ninth most common cancer in women in the United States.¹ Renal cell carcinoma (RCC, aka “kidney cancer”) is the most common cancer arising from the kidney. As guardians of the kidney, nephrologists have an increasing role in the management of patients diagnosed with RCC.² RCC was historically labeled as the “internist’s tumor,” a reference to the many presenting signs and paraneoplastic syndromes seen in patients with RCC, as well as to the limited role for surgery given the poor prognosis experienced by patients with advanced RCC. With detection of more early-stage tumors and improvement in surgical techniques and outcomes, patients with RCC are now primarily managed by urologists and interventional radiologists when the disease is localized to the kidney, as well as by medical oncologists when the disease is more advanced. Yet there is a resurgence of interest in the role of nephrologists in the care of patients with RCC, with some arguing that RCC should now be considered “the nephrologist’s tumor.”³ Nephrologists care

for patients with diseases associated with an increased risk of RCC. Nephrologists are also likely to identify incidental renal masses on imaging studies evaluating kidney disease and are integral to the preservation of kidney function in patients during and after RCC treatment. In this chapter, we discuss the epidemiology and biology of RCC, common treatments, and emerging research that is helping to better understand and treat patients with RCC.

EPIDEMIOLOGY OF RENAL CELL CARCINOMA

INCIDENCE OF RENAL CELL CARCINOMA

Kidney cancer can refer to any cancer arising in the kidney parenchyma or the renal pelvis. In the United States, kidney cancer accounts for approximately 4.0% of new cancers and 2.3% of cancer-specific deaths.¹ The vast majority (>90%) of kidney cancers are RCCs, such that the term “kidney

cancer” is often used to mean “RCC.” RCC accounts for 5% of cancers in men and 3% of cancers diagnosed in women worldwide. The incidence of RCC is increasing in both the United States and globally.^{4,6} The United States and the continent of North America are noted to have the highest relative incidence of kidney cancer,⁵ with a trend for lower incidence of RCC in less developed regions.

The incidence of RCC increases with age. In the United States, RCC presents most commonly in the sixth decade of life, while worldwide, most cases are diagnosed before 65 years of age.⁶ RCC is diagnosed in men at a rate two times higher than for women.^{7,8} The incidence of RCC also differs by designated race/ethnicity and genomic ancestry, with a slightly higher incidence of RCC among African Americans and Alaskan and Native Americans.^{8,9} These disparities in incidence persist, even accounting for known medical comorbidities associated with RCC.¹⁰

RENAL CELL CARCINOMA RISK FACTORS

There are many known risk factors for developing RCC, although each of these may explain only a fraction of the incidence of RCC.⁶

TOBACCO EXPOSURE

The association between smoking tobacco and RCC has been documented for many decades, as highlighted in Dr. Koop’s report from the Surgeon General in 1982.¹¹ Modern studies have continued to document that exposure to tobacco smoke is associated with an increase in RCC.¹² In a meta-analysis of 24 publications, the relative risk of RCC incidence was 1.31 higher in smokers compared with never-smokers, with the highest risk in heavy smokers.¹³ There appears to be a dose-dependent relationship between exposure to tobacco smoke and the risk of RCC,^{14,15} with attenuation of risk after smoking cessation.¹³ Smoking is also associated with an increased risk of presenting with higher-grade RCC, as well as higher RCC-related mortality.^{13,15}

OBESITY

Obesity and body weight have been associated with the risk of RCC in large national studies in Europe¹⁶ and systematic reviews.^{17,18} Some studies suggest this effect is most prominent for women.^{19,20} In China, metabolic syndrome was also associated with an increased risk of RCC.²¹ The association between obesity and RCC does appear to be dose dependent: The highest risk is among men and women with the highest body mass index (BMI). Obesity has also been associated with upstaging at the time of surgery²² and higher-grade RCC.²³ However, there remains an “obesity paradox,” where patients with obesity are at higher risk of developing RCC but may have better clinical outcomes.

HYPERTENSION

The association between high blood pressure (hypertension, HTN) and RCC was initially noted more than 30 years ago and appears to be stronger than for any other cancer type.²⁴⁻²⁶ Prospective studies in several countries confirmed that risk of RCC increased with presence of HTN in a dose-dependent manner, even when controlled for antihypertensive medications.²⁷⁻³¹ Furthermore, a recent study favored a role

of diastolic HTN rather than systolic HTN at the onset of RCC.³² However, any causal relation of HTN on RCC risk is difficult to disentangle from other conditions also associated with HTN and RCC (such as chronic kidney disease [CKD] and obesity) and has further complicated the evolving definition of HTN over the years.³³ Nevertheless, large observational studies that adjusted for these factors found HTN to be independently associated with RCC.¹⁶

OCCUPATIONAL AND ENVIRONMENTAL EXPOSURES

The geographic distribution of RCC cases suggests that there may be local factors including various environmental exposures that contribute to the risk of developing RCC. A meta-analysis of multiple studies suggested a direct association between air pollution and the risk of RCC, with a 5 $\mu\text{g}/\text{m}^3$ increase in fine particulate matter pollution ($\text{PM}_{2.5}$) associated with a 9% increased risk of RCC.³⁴ Serum measurements of perfluoroalkyl and polyfluoroalkyl substances (PFAS), which reflect environmental exposure to these pollutants, have also been associated with the development of RCC, particularly in White patients.^{35,36} Exposure to the industrial solvent trichloroethylene, either through occupational exposure or through contaminated water supplies, has also been associated with RCC.³⁷⁻³⁹ In addition, various occupations have been associated with RCC including firefighters, agriculture workers, dry cleaners, and metal workers.^{38,40,41} Occupational and environmental exposures are likely to increase with global urbanization, and knowledge of specific exposure risks can inform preventive measures in the future.

KIDNEY DISEASES ASSOCIATED WITH KIDNEY CANCER

KIDNEY CYSTS AND RENAL CELL CARCINOMA

Renal cysts, ranging from simple to complex, are considered ubiquitous, with estimates that more than 50% of adults older than 50 years of age will have at least one renal cyst on abdominal imaging.⁴² The Bosniak Classification defines several characteristics, based on computed tomography (CT) and MRI, that can help estimate the risk that a cyst harbors some RCC and provides guidelines for surveillance of more complex cysts^{43,44} (Table 43.1). In general, thickened septations within cysts, thickened or nodular cyst walls, and evidence of enhancement within these structures suggest an increased risk of cystic RCC. Simple cysts (Bosniak 1) do not require any additional follow-up or surveillance, while the majority of Bosniak 4 cysts contain RCC. While treatment and surveillance options for solid renal masses are discussed later in this chapter, it is important to note that Bosniak 3 and 4 renal cystic masses, like small solid renal masses, may be appropriate for surveillance strategies since these lesions are often associated with more indolent forms of RCC.⁴⁵

CYSTIC DISEASES AND RENAL CELL CARCINOMA

The link between cystic diseases and RCC was described in 1880 by Brigidi and Severi.^{46,47} CKD-related cystic diseases including acquired cystic kidney disease (ACKD) and polycystic kidney disease (PKD) have different characteristics from those of the general population and deserve further consideration.

Table 43.1 Bosniak Classification of Renal Cysts and Cystic Masses

Classification	Description	Historical Likelihood of Malignancy (%) ⁴⁴
Bosniak I	Simple cyst of water density with a thin wall that does not contain septa, calcifications, or other solid components	1.7
Bosniak II	Cyst may contain a few hairline septa and fine calcifications; uniform high-attenuation lesions that do not show contrast enhancement	18.5
Bosniak IIF	Cysts may contain multiple hairline thin septa or minimal smooth wall thickening	>18.5
Bosniak III	Thickened or irregular walls or septa with measurable enhancement	33
Bosniak IV	One or more enhancing nodules (>4 mm)	92.5

ACQUIRED CYSTIC KIDNEY DISEASE AND RENAL TUMORS

The development of ACKD has been described among 7% to 22% of patients with CKD, with higher proportions among patients with dialysis-dependent end-stage kidney disease (ESKD) (10%–44% within 1–3 years) and increases further with prolonged duration of dialysis (>90% after 5–10 years).⁴⁸ While patients with ACKD have historically been reported to have an elevated risk of malignant transformation to RCC,^{49,50} the predominant renal mass histology identified in patients with ACKD is clear cell papillary renal tumor, which is no longer listed as a malignant RCC in the 2022 World Health Organization (WHO) classification due to its indolent nature and low metastatic potential.⁵¹ Although the presence of ACKD leading to RCC has also been described among kidney transplant recipients (23%), the proportion of kidney transplant recipients with ACKD is far less than that observed in patients receiving maintenance dialysis (80%).⁴⁸ Acquired cysts seen in patients receiving maintenance dialysis have similar immunohistochemical profiles and histologic traits (epithelial cell-lined cysts with eosinophilic or foamy cytoplasm) when compared with tumors associated with ACKD, suggesting that acquired cysts are precursor lesions to renal tumors in this population.^{52,53} Future studies will be required to understand the optimal evaluation of clear cell papillary renal tumors in patients with ACKD, to determine optimal surveillance regimens, and to determine when treatment may be required.

POLYCYSTIC KIDNEY DISEASE AND RENAL CELL CARCINOMA

The prevalence of RCC in autosomal dominant polycystic kidney disease (ADPKD) does not appear to be greater than that in the general population, according to small case series and observational studies,^{50,54,55} although there is some controversy in the literature on this subject. As in the general population, clear cell RCC tends to be the most common histologic diagnosis of RCC in ADPKD, although in one European series, tubulopapillary pathology was prominently observed as well (42%).^{49,50,54} RCC in ADPKD presents at a younger mean age (50–60 years) than spontaneous RCC, but often with advanced disease, where one-third of patients have bilateral kidney involvement or metastatic disease. Presentation with advanced disease may be caused by delayed diagnosis, given the complexity of diagnostic images in the presence of multiple benign cysts in ADPKD. Symptomatic diagnosis rather than incidental discovery of RCC is more common in this select population.^{50,54}

CHRONIC KIDNEY DISEASE, END-STAGE KIDNEY DISEASE, AND RENAL CELL CARCINOMA

The causal relation between RCC and CKD is complex, with each pathology conferring risk to the other³ (Fig. 43.1). The reported 100-fold increased risk of RCC among patients with ESKD⁵⁶ may be due to the historical inclusion of clear cell papillary renal tumors in patients with ACKD, but this does not clearly explain the association of nondialysis-requiring CKD with RCC.⁵⁷ The largest cohort study to date has examined the risk of RCC due to CKD stratified by estimated glomerular filtration rate (eGFR) among 1,190,538 subjects without known RCC. With decreasing eGFR ranges, RCC risk increased as follows: stage 3a CKD, hazard ratio (HR) 1.39 (confidence interval [CI], 1.22–1.58); stage 3b, HR 1.81 (CI, 1.51–2.17); and stages 4 and 5, HR 2.28 (CI, 1.78–2.92).⁵⁷ The association was specifically with clear cell carcinoma and not with the other RCC subtypes. In contrast, the predominant RCC subtype seen in patients with ESKD is papillary RCC, suggesting pathology unrelated to ACKD. Moreover, findings of early or mild kidney pathology, such as albuminuria, have been related to cancer including RCC.⁵⁸ The link between kidney function and localized RCC risk was abundantly apparent among 202,195 kidney transplant recipients. During periods of graft failure resulting in ESKD, the incidence of RCC rose and, during periods of graft function, RCC incidence fell. This pattern recurred during periods of kidney function loss (repeat graft failure).⁵⁹ How loss of kidney function relates to RCC risk is largely unknown but may be related to increased use of imaging studies during evaluation of patients with impaired kidney function, or it may be attributed to inflammation, acidosis, or uremic factors often observed in advanced CKD.

BIOLOGY OF RENAL CELL CARCINOMA

CLASSIFICATION PRINCIPLES FOR RENAL CELL CARCINOMA

RCC includes a family of tumors all arising from the renal parenchyma. Clear cell RCC (ccRCC) is the most common, and the rest of the subtypes fall under the umbrella of non-clear cell RCC (nccRCC). The two most common nccRCC subtypes are papillary RCC (pRCC) and chromophobe RCC (chRCC). The WHO updated definitions of RCC subtypes in 2022, adding several new molecularly defined subtypes (Table 43.2).^{60,61} The development of RCC includes a

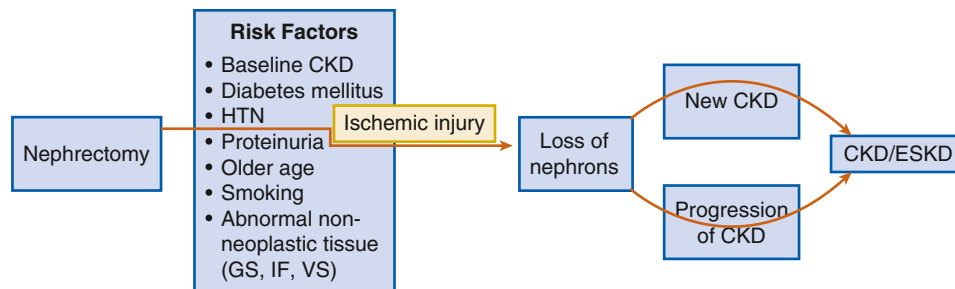


Fig. 43.1 Risk factors for chronic kidney disease (CKD). Nephrectomy performed for renal cell carcinoma is associated with nephron loss due to tissue removal, as well as ischemic and vascular injury. This loss of nephrons is associated with either new-onset CKD or an increased risk of progression of CKD in patients who have various risk factors, as noted. *ESKD*, End-stage kidney disease; *GS*, glomerulosclerosis; *IF*, interstitial fibrosis; *HTN*, hypertension; *VS*, vascular sclerosis.

Table 43.2 Subtypes of Renal Cell Tumors: World Health Organization Classification 2022

Subtype
Clear cell RCC
Papillary RCC
Oncocytoma
Chromophobe RCC
Collecting duct carcinoma
Medullary carcinoma, not otherwise specified
Acquired cystic disease–associated RCC
Eosinophilic solid and cystic RCC
Molecularly defined renal carcinomas:
TFE3-rearranged RCC
TFEB-altered RCC
ELOC (formerly TCEB1)-mutated RCC
Fumarate hydratase-deficient RCC
Succinate dehydrogenase-deficient RCC
ALK-rearranged RCC
SMARCB1-deficient medullary-like RCC
SMARCB1-deficient undifferentiated RCC, NOS
SMARCB1-deficient dedifferentiated RCC of other specific subtypes

RCC, Renal cell carcinoma.

complex interaction among patient risk factors, medical comorbidities, environmental exposures, and genetics.⁶² There is an increasing understanding of the genetic basis of RCC. The genetic or epigenetic loss of von Hippel-Lindau (VHL) tumor suppressor activity is a key event in the pathogenesis of ccRCC⁶³⁻⁶⁶ leading to decreased degradation of hypoxia-inducible factor (HIF), activation of hypoxia pathways, angiogenesis, and alterations in the tumor microenvironment.^{3,63,64,67-72} Recent advances in the genomic profiling of nccRCC have resulted in further characterization of tumor subtypes, allowing for better understanding of their biology and specificity of treatment targets.

RENAL CELL CARCINOMA HISTOLOGY AND GENETICS

The vast majority of RCCs are due to sporadic molecular events in kidney cells. However, an important subset of

RCCs arise due to germline mutations, manifesting at a much younger age (*VHL*: clear cell RCC; *MET*: papillary RCC; *FLCN*: chromophobe RCC; *TSC*: eosinophilic solid and cystic RCC, *FH*: FH-deficient RCC; *BAP1*: clear cell RCC; *SDHA/SDHB/SDHC/SDHD*: SDH-deficient RCC). The major subtypes of sporadic and germline RCCs are described as follows.

CLEAR CELL RENAL CELL CARCINOMA

Clear cell RCC is the most common type of kidney cancer and originates in proximal tubular cells. Genetic abnormalities in ccRCC are roughly divided into those affecting the *VHL* gene, epigenetic regulators and chromatin remodeling genes, and disruption of PI3K-AKT-mTOR signaling.⁷³⁻⁷⁵ The *VHL* gene has three exons and is located on the short arm of chromosome 3 (3p25-26).⁷⁶ Sporadic ccRCCs are characterized by loss of VHL function through loss of the short arm of chromosome 3, deletions, mutations, or epigenetic silencing of the *VHL* gene.^{67-69,77-80}

The *VHL* gene plays a crucial role in a variety of cellular pathways. The key function of VHL protein (pVHL) is to serve as the substrate recognition unit in a protein complex responsible for polyubiquitination of a variety of proteins, including the HIF transcription factor complex.⁸¹ Polyubiquitination marks HIF for degradation in the presence of normal tissue oxygenation (normoxia).⁸² HIF protein subunits are encoded by *HIF1A*, *EPAS1*, *HIF3A*, *ARNT*, *ARNT2*, and *ARNT3* genes. HIF1-alpha (HIF1a) or HIF2-alpha (HIF2a), which are the most important subunits in the pathogenesis of ccRCC, heterodimerize with HIF1-beta (HIF1b) to activate the transcription of hypoxia-inducible genes, such as vascular endothelial growth factor (VEGF), and to upregulate the VEGF receptor (VEGFR). The resulting clinical phenotype is characterized by florid vasculature, principally modulated by VEGF signaling pathways.⁸³ In addition, HIF increases transcription of erythropoietin, platelet-derived growth factor-β (PDGF-β), transforming growth factor-α (TGF-α), and several glycolytic enzymes.⁸⁴⁻⁸⁶ Failure of pVHL to rein in the activity of HIF in normoxia has crucial implications for the clinical behavior and metabolic features of ccRCC. Among them are erythropoietin-dependent erythrocytosis, marked angiogenesis, and cell proliferation mediated by TGF-α and VEGF,^{84,87} as well as aerobic glycolysis (Warburg effect) and the immunosuppressive effects of increased tryptophan catabolism.⁷¹

Sporadic ccRCC commonly has additional mutations in epigenetic regulators, also localized to chromosome 3p25. These include polybromo 1 (*PBRM1*), BRCA-associated protein-1 (*BAP1*), and SET domain-containing 2 (*SETD2*) genes involved in regulation of chromatin maintenance and remodeling. Generally, hypermethylation of promoter sites is associated with the tumors carrying these mutations (except for *SETD2*) and higher tumor grade.⁸⁸⁻⁹² Loss of these genes further increases hypoxia signaling and genomic instability to fuel the RCC neoplastic phenotype.^{80,93}

Genes encoding components of the mammalian target of rapamycin (mTOR) signaling pathway, such as *PTEN*, *AKT*, *PIK3CA*, and *MTOR* genes, are involved in cell growth, and cell proliferation signaling. Aberrations in these genes have been identified in up to 28% of RCC patients.⁸⁸⁻⁹² *PTEN* deficiency in particular is associated with more aggressive tumors.⁹⁴ Together with the abnormal expression of the focal adhesion kinase modulator of the mTOR pathway, these genetic defects are present in more than 50% of ccRCC patients.⁶⁴

The majority of ccRCC cases are sporadic; only 2% to 3% of ccRCC cases are linked to known hereditary diseases. Germline mutations in the *VHL* gene⁹⁵ lead to VHL disease, an inherited syndrome with an estimated incidence of 1 in 36,000 live births, and a penetrance of more than 90% by age 65 years.^{80,93} Most subjects affected by VHL disease inherit a germline mutation of the gene from the affected parent and a normal gene from the unaffected parent. All the cells of affected individuals who inherit the genetic trait will have germline mutations in *VHL*. However, tumors will develop only in those cells that undergo functional inactivation of the remaining wild-type allele (the second hit) and, in addition, are constituents of vulnerable target organs, with the formation of cysts, benign or malignant tumors in kidneys, the central nervous system, adrenal glands, pancreas, and reproductive adnexal organs. Germline *BAP1* mutations can also be a cause of an autosomal dominant inherited predisposition for ccRCC.^{96,97} When germline familial syndromes are present, careful monitoring for renal lesions starts at young ages, guided by expert genetics teams.

PAPILLARY RENAL CELL CARCINOMA

Papillary RCC (pRCC), also of proximal tubule origin, is the most common nccRCC.⁹⁸ Previously, pRCC was divided into type 1 (typically characterized by *MET* mutations) and type 2 (which included RCCs that were fumarate hydratase [FH]-deficient or *SMARCB1*-deficient). In the latest 2022 WHO Classification, the type 1 and type 2 categories were removed because most pRCC tumors have both type 1 and type 2 histologic features and both types were determined to be composed of multiple molecular subtypes^{60,93,99} (Table 43.2). The current pRCC category still includes some patients who have a hereditary form (e.g., germline *MET* proto-oncogene-activating mutations). Therefore it is recommended that all new diagnoses of pRCC undergo genetic testing, especially if multifocal lesions are present.¹⁰⁰ *MET* mutations can also occur in 10% to 20% of cases of somatic mutations in the sporadic form of pRCC.¹⁰¹ Additionally, trisomy 7 has been implicated in *MET* amplification in some cases of pRCC.¹⁰² Mutations in the epigenetic modifier genes *PBRM1*, *BAP1*, and *SETD2* have also been described with this tumor type but seem to be less prevalent compared with ccRCC.¹⁰² As

additional molecular subtypes continue to be elucidated, it is possible that a subset of tumors currently classified as pRCC may, in turn, be renamed as distinct subtypes.

CHROMOPHOBE RENAL CELL CARCINOMA

Comprising less than 5% of RCCs, this rare nccRCC originates from the renal collecting duct. ChRCC has a histologic appearance similar to benign oncocytomas, which can make diagnosis between these two distinct conditions challenging, especially when examining biopsy specimens.¹⁰³ ChRCC is often linked with whole-chromosome losses and germline mutations in the folliculin gene (*FLCN*) found in the autosomal dominant Birt-Hogg-Dubé syndrome.¹⁰⁴ The most frequent mutation in sporadic cases of chRCC involves downregulation of p53 tumor suppressor signaling and loss of function of *PTEN* leading to phosphatidylinositol-3 kinase (PI3K)-driven cell proliferation. ChRCC is more common in younger females and is typically the least aggressive of all RCC types, unless characterized by sarcomatoid transformation or metastasis.¹⁰⁵

MOLECULARLY DEFINED RENAL CELL CARCINOMA SUBTYPES

Two rare nccRCCs are notable for their association with germline mutations and distinct clinical presentations and treatment options: FH-deficient RCC is one of the most aggressive RCCs.¹⁰⁶⁻¹⁰⁸ Loss of FH function can be a consequence of either germline or somatic alterations in the *FH* gene. Germline losses of *FH* lead to development of RCC at a very young age, thus screening for renal masses starts as early as age 8.¹⁰⁹ FH loss can also be part of a genetic syndrome called hereditary leiomyomatosis and renal cell cancer (HLRCC).¹¹⁰⁻¹¹⁴ HLRCC occurs predominantly in females and is associated with uterine fibroids. Another extremely rare RCC is *SMARCB1*-deficient RCC, associated with sickle cell trait and germline mutations in the *SMARCB1* gene.¹¹⁵ Treatment of FH-deficient RCC and *SMARCB1*-deficient RCC differ from ccRCC, pRCC, and chRCC, as discussed in “Systemic Therapy” section later.

CLINICAL PRESENTATION OF PATIENTS WITH RENAL MASSES

The unusual and pleiomorphic presentation of RCC is reflective of its biologic characteristics, with clinical presentation unlike that of most other urologic tumors, and this challenge is reflected in its well-deserved appellation “the internist’s tumor.” Thus the diagnosis can be challenging due to its nonspecific and often systemic symptoms.

The clinical presentation of RCC is variable; roughly 15% of patients have metastatic disease at the time of presentation.¹ The classic presentation of hematuria, flank pain, and fever is seen in less than 10% of patients,^{116,117} and more than 50% of patients are now diagnosed on the basis of incidental findings in abdominal radiologic imaging performed for unrelated reasons.¹¹⁸ The early detection of RCC has led to fewer patients presenting with paraneoplastic syndromes; however, historically several nonspecific clinical findings including hypertension (HTN), weight loss, malaise, night sweats, and the new onset of a varicocele were more common.

Paraneoplastic syndromes including fever, anemia, hypercalcemia, erythrocytosis, and abnormal liver enzyme levels not due to metastatic spread (Stauffer syndrome) are seen in 7% of cases; polyneuropathy and amyloid A amyloidosis occur in 3% to 5%.¹¹⁶ Fever occurs in up to 20% of patients with paraneoplastic syndromes and is often accompanied by night sweats, anorexia, weight loss, and fatigue.¹¹⁶ Hypercalcemia occurs in up to 15% of metastatic cases and has been identified as a negative prognostic factor, sometimes associated with lytic bone metastases.¹¹⁹ The ectopic production of parathyroid-related peptide by the tumor has been linked to hypercalcemia in the absence of bone metastases.^{120,121} Erythrocytosis has been described in 1% to 5% of patients with RCC and is thought to be secondary to the unregulated production of erythropoietin by cancer cells.¹²²

EVALUATION AND DIAGNOSIS OF RENAL MASSES

The incidence of RCC has been increasing worldwide.^{5,6} This increasing incidence has been driven in part by the increasing use of modern abdominal imaging techniques and the incidental detection of small renal masses. The rate of incidental diagnosis increased from about 10% in the 1970s to 60% fewer than 3 decades later. Given the frequent use of imaging to evaluate the kidneys, nephrologists may be the first to identify renal masses. As a consequence, there has been a profound stage migration, such that most patients are now diagnosed with early-stage RCC. There are currently no screening tests for early detection of RCC. Although imaging, blood, and urine biomarkers for early detection of RCC are areas of active research, there are significant barriers. For example, compared with many other cancer types, RCC tumors seem not to shed a lot of material into the bloodstream (e.g., DNA).¹²³⁻¹²⁶ Like most cancers, the goal is to detect early-stage, curable RCC lesions so that they can be successfully treated to prevent progression to incurable, advanced RCC.

RADIOLOGIC EVALUATION OF RENAL MASSES

Along with clinical assessment, diagnostic imaging has become the mainstay of renal mass evaluation. The primary choices of imaging modality for the evaluation and staging of renal masses are CT or MRI.¹²⁷⁻¹²⁹ Current guidelines recommend the use of CT abdomen with intravenous contrast as the primary imaging modality to evaluate a renal mass in patients with no contraindication to use of contrast.^{100,130}

MRI is traditionally considered to be optimal for evaluating soft tissue pathology and, in the case of RCC, can provide excellent imaging of microscopic and macroscopic fat and intracystic structures, as well as for evaluation for tumor thrombus presentation in the inferior vena cava (with 81%–86% accuracy for T-staging).¹²⁹ Although MRI remains highly effective in renal mass characterization and staging, the newer multidetector CT scanners have achieved an equivalent level of accuracy and detection rate (with a sensitivity of 59% to 88% and a specificity of 71% to 93% in detecting stage T3a RCC) as compared with MRI

in most scenarios. Therefore ultimately, CT has become the primary imaging modality to aid in the management of RCC.¹³¹

Ultrasound is also commonly performed during routine medical care, which has led to incidental findings of renal masses. It is particularly useful for distinguishing noncystic tumors from simple and complex cysts. However, current guidelines recommend CT and MRI over ultrasound due to the inability of ultrasound to evaluate for locally advanced or metastatic disease.¹³² Use of ultrasound can also be limited for patients with a larger body habitus. However, ultrasound is now routinely used during laparoscopic and robotic surgery to identify renal masses before resection.¹³³ Once the lesion is identified, ultrasound can be used for repeated monitoring, especially when avoidance of iodinated intravenous contrast is preferable.¹²⁷ Novel contrast-enhanced ultrasound using microbubbles of injectable gas has been found to be helpful in indeterminate renal masses and in patients who are poor candidates for CT or MRI contrast agents.^{134,135}

Positron emission tomography (PET)–based imaging techniques are emerging tools for evaluating and staging RCC. The ¹⁸F-fluoro-2-deoxy-2-d-glucose (FDG) PET tracer is commonly used in many malignancies but has more limited utility in RCC, as FDG is excreted by the kidneys, resulting in difficulty discriminating renal tumors from normal parenchyma. Sensitivity is as low as 32% for primary RCC kidney lesions.^{136,137} Due to the variability in FDG uptake in ccRCC, additional PET agents are in development: Girentuximab is an antibody that targets carbonic anhydrase 9 (CA9) on the membranes of RCC cells.^{138,139} A novel probe targeting CA9 using zirconium, ⁸⁹Zr-DFO-girentuximab, recently demonstrated 86% sensitivity and 87% specificity to detect early-stage ccRCC in a prospective study.¹⁴⁰ Additional probes targeting the RGD motif of α -v- β -3 integrin on endothelial cells,¹⁴¹ a glutamine analog 4-¹⁸F-(2S,4R)-fluoroglutamine,¹⁴² and a glutamic acid probe (S)-4-(3-[¹⁸F]-fluoropropyl)-l-glutamic acid [¹⁸F-FSPG] have all been shown to have uptake in patients with ccRCC.¹⁴³ PET-based imaging has future potential for RCC diagnosis and staging, patient selection, and monitoring of therapeutic response.

DIFFERENTIATION OF BENIGN TUMORS FROM RENAL CELL CARCINOMA

RCC holds a distinction as being one of the few cancer types that commonly undergoes surgical treatment without preceding histologic confirmation (i.e., a renal mass biopsy) of a cancer diagnosis. While imaging modalities are sensitive, and masses that are concerning for kidney cancer are confirmed to be cancer in the majority of cases, there remains a concern for overtreatment of benign renal masses.¹⁴⁴ Some 15% to 25% of patients treated with surgery to remove a small renal mass were ultimately found to have a benign renal mass.^{145,146} Yet historically, only one in five patients with a renal mass underwent a renal mass biopsy before treatment.¹⁴⁷ There is now an increasing interest in the use of renal mass biopsy to help distinguish benign renal masses from RCC.

The most common benign renal lesions are angiomyolipomas (AMLs), which are composed of blood, smooth

muscle, and adipose tissues. They are primarily seen among middle-aged women, 40 to 60 years of age.¹⁴⁸ Angiomyolipomas tend to be adipose-rich tumors that are more easily distinguished from RCC on imaging. Angiomyolipomas have a typical hyperechoic appearance on ultrasound and demonstrate macroscopic fat within the lesion on CT or MRI.

Oncocytomas are the next most frequently found benign renal mass, generally seen among older men (aged in their 70s). These tumors are derived from tubular cells in the cortical collecting duct and can be difficult to differentiate from RCC on imaging. Oncocytoma is difficult to distinguish from chromophobe RCC with a renal mass biopsy, as both have an oncocytic appearance; larger pathology samples are therefore usually needed to determine distinguishing characteristic architecture.¹⁰³ As a result, most renal mass biopsy specimens thought to be oncocytomas are classified as an “oncocytic neoplasm” to account for the diagnostic uncertainty for either oncocytoma or chRCC.

When having microscopic confirmation of an RCC diagnosis would change management or when a renal mass is indeterminate on routine imaging (although carbonic anhydrase-based PET has the potential to aid in diagnosis in the future; see previously), percutaneous renal biopsy can be safely performed with minimal risk of seeding (0.01% of cases). One series of 2474 percutaneous biopsies¹⁴⁹ reported good performance with respect to sensitivity (80%–100%), specificity (83%–100%), negative predictive value (82%), and positive predictive value (97.5%).^{148,150} Use of kidney biopsy is likely to increase in the coming years as molecular markers may help personalize treatment decisions.

TUMOR STAGING AND GRADING

Tumor-node-metastasis (TNM) staging for RCC was first established in 1997 by the Union Internationale Contre le Cancer (UICC) and American Joint Committee on Cancer (AJCC) and was most recently updated in 2017. The T-staging (T0–T4) reflects the size of the tumor and whether the tumor invades local structures (e.g., perinephric fat). T1a includes tumors ≤ 4 cm in size; T1b includes tumors 4–7 cm; T2 tumors are >7 cm but still confined to the kidney; T3 tumors invade major veins or perinephric tissues while T4 tumors extend beyond Gerota’s fascia and may include the ipsilateral adrenal gland. The N stage (N0/N1) reflects whether regional lymph nodes are involved. The M stage (M0/M1) describes whether there are distant metastases. Composite prognostic staging (I–IV) summarizes TNM findings (Fig. 43.2). The TNM staging is used in two different situations: 1. Clinical TNM staging reflects the characteristics of the tumor using findings on imaging and examination, and 2. Pathologic TNM staging reflects information from pathologic examination.

The tumor grade of RCC is based on the microscopic appearance of nuclear features (e.g., nuclear size and nucleolar prominence). Current grading systems for clear cell and papillary RCC use the International Society of Urologic Pathology (ISUP)-defined grading system (Fig. 43.3). This ISUP grading system was adopted by the World Health Organization and is often referred to as the WHO/ISUP grading classification.¹⁵¹ In addition to nuclear features, the dedifferentiation of high-grade RCC epithelial cells can result in sarcomatoid or rhabdoid morphology, a poor prognostic feature.¹⁵²

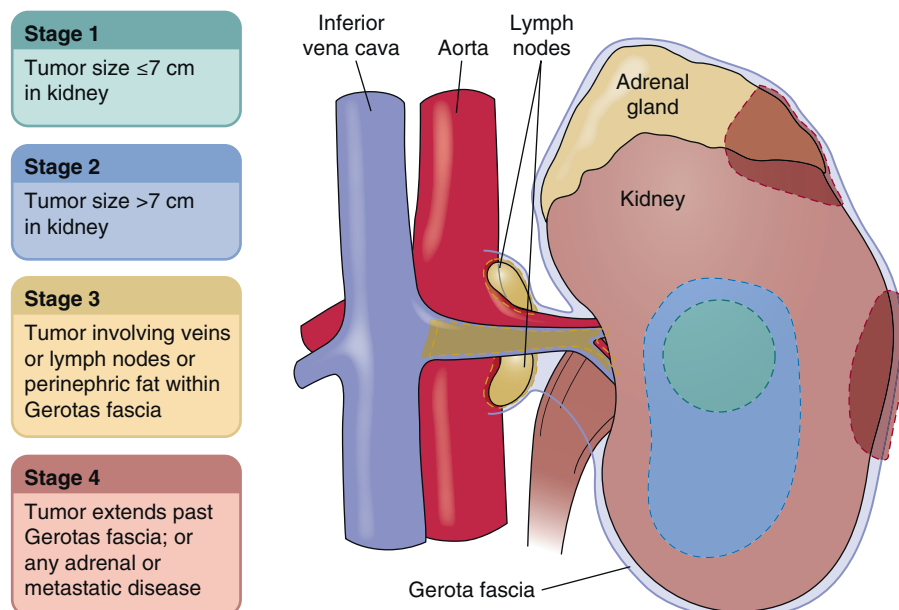


Fig. 43.2 The tumor-node-metastasis (TNM) staging classification. Shown is the general staging by the American Joint Committee on Cancer (AJCC) and Union for International Cancer Control (UICC), updated in 2017. (From Amin MB, Edge S, Greene F, et al., eds. *AJCC Cancer Staging Manual*. 8th ed. Springer International Publishing: American Joint Commission on Cancer; 2017 and Brierley JD, Gospodarowicz MK, Wittekind C, eds. *TNM Classification of Malignant Tumours*, 8th ed. Wiley-Blackwell; 2017.)

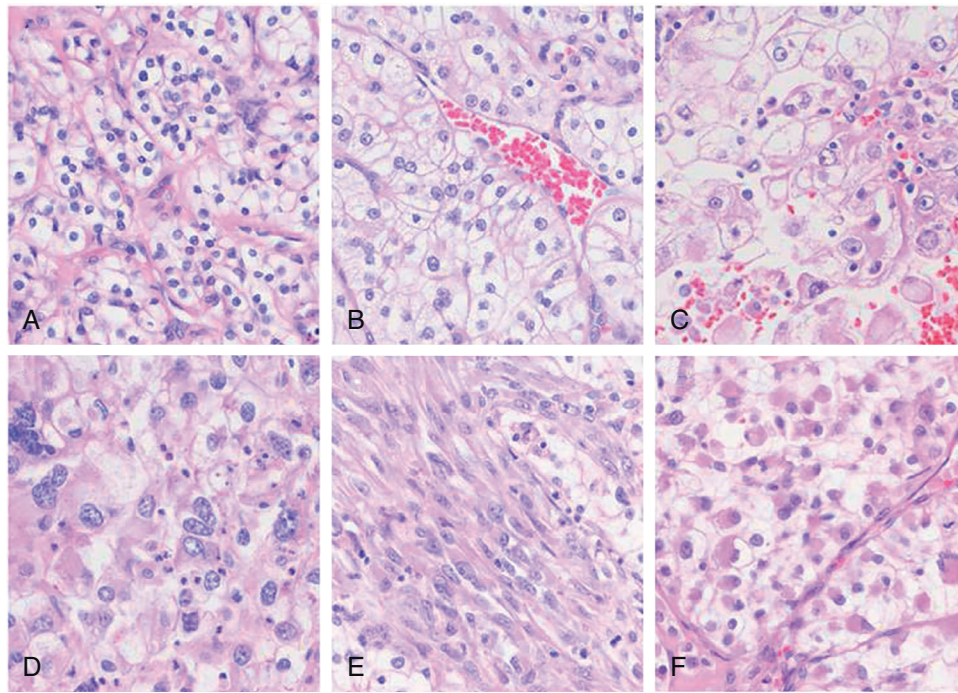


Fig. 43.3 The International Society of Urologic Pathology (ISUP) grading system for clear cell and papillary renal cell carcinoma (RCC). (A) Grade 1: Nucleoli are inconspicuous or absent. (B) Grade 2: Nucleoli are clearly visible at high-power magnification but are not prominent. (C) Grade 3: Nucleoli are prominent and are easily visualized at low-power magnification. (D) Grade 4: Presence of tumor giant cells and/or marked nuclear pleomorphism. (E) Grade 4: Sarcomatoid carcinoma. (F) Grade 4: Carcinoma showing rhabdoid differentiation.

PROGNOSIS AND CANCER OUTCOMES

PROGNOSIS OF RENAL CELL CARCINOMA IN THE GENERAL POPULATION

Outcomes for patients diagnosed with RCC span a wide spectrum. Patients with localized RCC experience excellent survival outcomes (more than 92% cancer free at 5 years after treatment).¹⁵³ In contrast, the 5-year relative survival for patients with metastatic RCC is historically dismal (15%, according to the SEER database from 2013–2019).¹⁵³ Treatment of patients diagnosed with locally advanced and metastatic RCC has benefited from the introduction of many new systemic therapies in the past 20 years. Modern clinical trials have reported significantly improved survival outcomes. In the recent phase III study of front-line immunotherapy for patients with metastatic RCC, 50% of patients survived 55 months, highlighting how access to optimal care can positively affect outcomes.¹⁵⁴

Various integrated staging systems have been proposed to provide more personalized prognostic information for patients and clinicians. These incorporate TNM staging, tumor size, clinical symptoms, histologic subtype, grading, and other prognostic pathologic findings that can be used to provide prognostic information for patients and clinicians.^{155–157} For patients with metastatic RCC, the Memorial Sloan Kettering Cancer Center (MSKCC) criteria and International Metastatic Renal Cell Carcinoma Database Consortium (IMDC) criteria use select laboratory test results and patient performance status to risk stratify patients into low-, intermediate-, and high-risk categories and to help select optimal treatment

regimens.^{119,158,159} Other systems have proposed expanding the number of clinical laboratory tests used in models to improve risk stratification.¹⁶⁰ Future models are likely to include machine learning and artificial intelligence to extract prognostic features from clinical imaging and from electronic health records to further improve prognostication.

PROGNOSIS OF RENAL CELL CARCINOMA IN THE END-STAGE KIDNEY DISEASE AND TRANSPLANT POPULATIONS

The prognosis of RCC in the ESKD population is equivalent or better compared with that in the general population.^{161,162} The 5-year survival from renal tumors related to ACKD was comparable between the two groups.⁵⁵ RCC in transplant recipients tends to have favorable characteristics—younger age, smaller tumor size, less metastatic disease, more stage T1a tumors, and papillary subtype diagnosis—when compared with persons with ESKD receiving dialysis.¹⁶³

Survival outcomes for transplant recipients with RCC have varied between studies but generally appear to be more favorable compared with those of the general population, and possibly the dialysis population, likely due to closer surveillance of these patients. The 10-year cancer-specific survival of 88% to 95% in transplant recipients is higher than that in the general population (75%).¹⁶⁴ Additionally, 97% of transplant recipients achieved a cancer-specific survival of 5 years, better than 77% in the nontransplanted ESKD group in one study.¹⁶⁵ The overall patient and graft survival between those with and without RCC have been comparable.¹⁶³

TREATMENT OF LOCALIZED AND LOCALLY ADVANCED RENAL CELL CARCINOMA

Treatment of RCC diverges into two arms depending grossly on whether the tumor is localized to the kidney (with or without local invasion or regional lymph node involvement) or has distant spread. Surgery remains the mainstay treatment for patients diagnosed with localized or locally advanced RCC, whereas systemic therapies are the mainstay of treatment for patients diagnosed with distant metastatic disease.

SURGICAL TREATMENT

PREOPERATIVE CLINICAL EVALUATION

In addition to common considerations about surgical candidacy, guidelines suggest that patients with localized or locally advanced RCC considering surgery should also be assessed for their risk of significant CKD or ESKD following surgery.^{100,166,167} This prognostic information can aid patients and clinicians in deciding between surveillance and treatment or choosing the type of surgical treatment to pursue. Identification of preexisting CKD (serum creatinine level and eGFR calculation) and proteinuria assessment (urine albumin-to-creatinine or urine protein-to-creatinine ratio) further allows for CKD staging using the Kidney Disease Improving Global Outcomes (KDIGO) classification criteria.¹⁶⁸ Similarly, the risk of ESKD can be estimated by the Kidney Cancer Risk Equation, adapted from the Kidney Failure Risk Equation (KFRE).¹⁶⁹ Further characterization of differential kidney function with renal nuclear scintigraphy before nephrectomy may help prognosticate the risk of CKD after treatment; however, these studies are limited because they tend to underestimate the postoperative function of the remaining kidney, which is often modestly augmented by compensatory hyperfiltration and hypertrophy.^{170,171} Preservation of kidney function may be achieved by addressing CKD risk factors before RCC intervention. Optimization of comorbid diseases encompassing glycemic control among diabetics, blood pressure adjustment for those with HTN, prevention of acute kidney injury through medication review, and avoidance of hypotension and nephrotoxicity have been recommended.

RENAL MASS EXCISION AND NEPHRON-SPARING THERAPIES

For decades since the first nephrectomy in 1861, the primary treatment of kidney cancers had been radical nephrectomy, which involved total removal of the kidney and surrounding tissues, such as the adrenal glands or lymph nodes. Radical nephrectomy remains the primary treatment for locally advanced RCC and is increasingly supported by adjuvant systemic therapies for patients at a high risk of recurrence, with neoadjuvant treatment regimens on the horizon (discussed further later). Nephron-sparing surgery (i.e., partial nephrectomy) was originally reserved for imperative indications such as a renal mass in a solitary kidney. With the increase in appreciation for the importance of preservation of parenchymal mass and renal function, elective partial nephrectomy for eligible localized renal masses gained

popularity. Partial nephrectomy has demonstrated equivalent oncologic outcomes for patients with small renal masses.¹⁷² The widespread adoption of technologies like robotic surgery has further encouraged the utilization of partial nephrectomy.

Partial nephrectomy has been associated with improved renal function preservation when compared with radical nephrectomy¹⁷³ and, importantly, has also been associated with improved survival outcomes.¹⁷⁴⁻¹⁷⁶ A meta-analysis of 40,000 subjects during the early adoption of elective partial nephrectomy demonstrated improved all-cause and cancer-specific mortality for patients receiving partial nephrectomy.¹⁷⁷ However, one randomized controlled trial, the European Organization for Research and Treatment of Cancer (EORTC) study, which assessed 541 patients with renal masses 5 cm or smaller, did not demonstrate improved survival with partial nephrectomy.¹⁷⁸ This likely illustrates the selection bias that younger and healthier patients are more likely to receive a partial nephrectomy.¹⁷⁹ With increasing access to experienced surgeons and minimally invasive techniques, the safety of both radical and partial nephrectomy has continued to improve.¹⁸⁰ Currently, partial nephrectomy (when technically feasible) is a guideline-recommended treatment for patients with small renal tumors, Bosniak 3 or 4 renal cysts, baseline CKD, risk factors for progressive CKD following surgery, and multifocal tumors.^{166,167}

ABLATIVE THERAPIES

Ablative therapies including percutaneous cryoablation and radiofrequency ablation may also be considered as a primary treatment option for small renal masses (<3 cm) and patients considered to be poor surgical candidates.^{100,166,167} Percutaneous ablation procedures (typically performed by interventional radiologists using CT guidance) are well-established nephron-sparing procedures that may be preferred for older adults and those with multiple comorbid conditions, in part because of the quicker recovery and avoiding the need for general anesthesia.¹⁸¹ These less invasive ablative therapies have the advantage of similar oncologic survival (in select cases), lower procedural complication rates, and fewer hospitalization days.¹⁸²

SURVEILLANCE

Active surveillance entails monitoring of the renal mass with imaging. Surgical intervention is prompted by concerning growth over time¹⁸³ or other changes, as indicated.^{184,185} Compared with more aggressive measures, which start with surgical or ablative treatment, active surveillance performs well in select populations and those with lower life expectancy or competing risks of death.¹⁸⁶ The American Urological Association guidelines have recommended active surveillance for those with masses 2 cm or smaller or with high operative risk.^{166,167}

POSTTREATMENT SURVEILLANCE OF LOCALIZED OR LOCALLY ADVANCED TUMORS

The American Urological Association has recommended regular screening for tumor recurrence by imaging (abdominal CT, MRI, or ultrasound and chest radiography)

initially after treatment. Imaging is performed on the basis of the risk of recurrence, with annual imaging recommended for low-risk tumors and imaging every 3 to 6 months for tumors at a high risk of recurrence.^{166,167}

SYSTEMIC THERAPY FOR RENAL CELL CARCINOMA

INTRODUCTION TO SYSTEMIC THERAPIES

Although incidental renal mass detection has led to earlier RCC diagnosis, with favorable outcomes, up to one third of RCC patients are diagnosed with advanced, metastatic, or recurrent disease that is not amenable to surgical therapy.⁵ For these patients, systemic therapy is the mainstay of treatment, although surgical debulking may be indicated for reducing tumor burden in some cases.¹⁸⁷⁻¹⁸⁹ Initial systemic therapeutic approaches in the 1980s to 1990s exploited the high level of immunogenicity of RCC and used immunotherapy with interferon or interleukin-2 (IL-2), but these were associated with severe toxicity and only modest success. Since that time, remarkable advances have been made, leading to U.S. Food and Drug Administration (FDA) approval of several tyrosine kinase inhibitors (TKIs) of VEGFR2 (sunitinib, pazopanib, sorafenib, axitinib, cabozantinib, lenvatinib, and tivozanib), an antibody targeting VEGF (bevacizumab), a small molecule HIF2a inhibitor (belzutifan), and mTOR kinase inhibitors (temsirolimus, everolimus)¹⁹⁰⁻²⁰³ (Fig. 43.4). The arrival of immune checkpoint inhibitors (ICIs) has dramatically improved overall survival (OS) of ccRCC,²⁰⁴ alone and in doublet combinations. In the current era, first-line treatment is generally a doublet combination therapy: an ICIs plus a VEGFR inhibitor, or a combination of two immune checkpoint inhibitors.^{154,205-209} With these combinations, median survival for patients on clinical trials of advanced RCC has improved to 49.5 to 55 months.¹⁵⁴ In the second and later line settings, systemic therapy with single-agent TKIs or HIF2a inhibitors is the mainstay of treatment. One-year survival for advanced RCC has improved from 39.1% in 2009 to 49% in 2019. As newer therapies and combinations are developed, patients with advanced RCC are expected to show continuing increases in OS.²¹⁰

IMMUNOTHERAPY

CYTOKINE THERAPY

Interferon- α (IFN- α) and IL-2 were the initial immune therapies to be approved by the FDA. Cytokine therapy stimulates the immune system nonspecifically with the upregulation of T-cell and natural killer cell activity to recognize and destroy tumor cells,^{211,212} but it can also have the undesired effect of suppressing the immune system and defeating natural immunosurveillance.^{213,214} Response rates ranged from 10% to 25% and median OS from 11 to 13 months, with 5% sustained complete response for highly selected patients. Cytokine therapy, especially high-dose IL-2, is extremely limited by hypotension and “flu-like” adverse events and could only be offered to patients with excellent heart, lung, and renal function.^{215,216}

IMMUNE CHECKPOINT INHIBITOR THERAPY

Immune checkpoint inhibitors (ICIs) have made immunotherapy far more accessible to patients and have nearly completely replaced cytokine therapy due to their higher tolerability and effectiveness. Immune checkpoint receptors sustain normal autoregulation of our immune system by downregulating T-cell and associated immune cell (macrophage, natural killer cell) activity. Tumors cells “hijack” this function to evade circulating immune cells. ICIs block the aberrant tumor-to-T-cell interaction, enabling T-cells to recognize the deviant cells and stimulate the natural immune mechanisms to eradicate them. Cytotoxic-T lymphocyte-associated antigen-4 (CTLA-4) and programmed cell death protein-1 (PD-1) are two receptor proteins regulating such adaptive immunity used in cancer treatment. Programmed cell death-ligand 1 (PD-L1) is the ligand for PD-1. Ipilimumab (anti-CTLA-4 antibody) and nivolumab (anti-PD-1 antibody) have transformed the outlook of patients with RCC.^{154,217} Single-agent nivolumab can shrink a minority of tumors but does increase OS, whereas single-agent ipilimumab has low activity.^{204,218} However, a front-line combination regimen of ipilimumab with nivolumab is synergistic in patients with advanced ccRCC, yielding an overall response rate of 42% including 9% complete response rate and an excellent median OS of 53 months.²⁰⁵ Furthermore, for patients who have a response to this combination, the median duration of response is an impressive 75 months.²¹⁹ Most strikingly, patients with sarcomatoid or rhabdoid histologies, historically with the worst outcomes, benefit the most from checkpoint inhibition.²²⁰ In most patients with metastatic ccRCC, it is standard of care to use combination ICI, typically in first-line therapy, or monotherapy in a later line of therapy, if not received in first line.

Toxicity related to checkpoint inhibitors tends to present with cutaneous, gastrointestinal, pulmonary, and hepatic involvement. Less commonly, renal complications can occur, with acute kidney injury due to acute tubulointerstitial nephritis (lymphocytic infiltration), which typically resolves with corticosteroid administration or cessation of treatment (as detailed in Chapter 57).^{221,222}

Use of ICIs must be carefully considered in patients with preexisting autoimmune conditions or renal transplants, as immune tolerance can be broken. Retrospective studies demonstrate an approximately 40% graft rejection rate including a retrospective study of 69 patients by the Immune Checkpoint Inhibitors in Solid Organ Transplant Consortium that demonstrated 42% of patients developed acute rejection and 28% experienced loss of allograft, at a median time of 24 days after first dose of ICI.²²³ However, in a prospective phase I study of 17 patients who remained on immunosuppression while receiving ICI for various cancers, rejection episodes attributed to the ICI occurred in only two patients (12%) and one had reversal of rejection through use of plasma exchange. Importantly, despite ongoing immunosuppression, 53% of patients had tumor responses to the ICI, including 24% who had a complete response.²²⁴ In patients with renal transplants, ICI treatment can be administered if the need for tumor control outweighs the risk of graft rejection. For patients on dialysis, because antibodies are cleared through the hepatic system, ICIs may be given and they result in excellent efficacy.^{225,226}

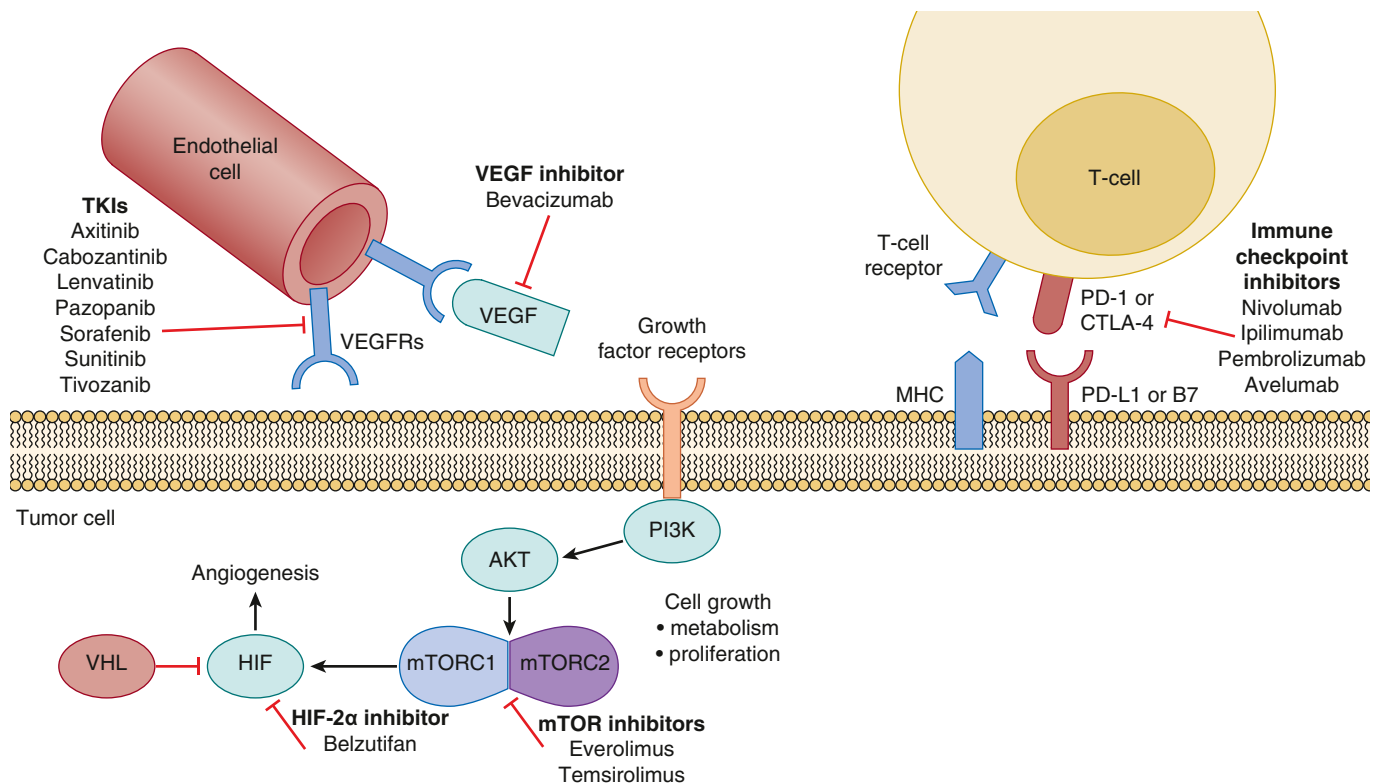


Fig. 43.4 Tumor growth, angiogenesis, immune pathways, and associated targeted therapy. Tumor cells promote growth and angiogenesis through the PI3K/mTOR and VEGF pathways and evade the T-cells and immune checkpoints. Various targeted immune therapies are listed; the site of action is indicated by the red line. B7, CD-28 ligand; CTLA-4, cytotoxic-T-lymphocyte-associated antigen-4; HIF, hypoxia-inducible factor; MHC, major histocompatibility complex; MTOR, mammalian target of rapamycin; mTORC, mammalian target of rapamycin complex; PD-1, programmed cell death protein-1; PD-L1, programmed death-ligand 1; PI3K, phosphatidylinositol-3-kinase; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor; VEGFR, VEGF receptor; VHL, von Hippel-Lindau.

Resistance to ICIs can be roughly grouped into tumor intrinsic and tumor extrinsic mechanisms. Tumor intrinsic mechanisms include decreased antigen presentation and decreased interferon gamma signaling.²²⁷⁻²³¹ Beyond molecular characteristics found within tumor cells, extrinsic mechanisms such as host factors may also modify response to ICIs. For example, use of antibiotics can alter ICI efficacy and an individual's microbiome may also confer sensitivity or resistance to ICIs.²³²⁻²⁴⁰ There appears to be cross-resistance among ICIs in RCC, which is borne out clinically: after progression with front-line ICI, there is limited value in either switching ICIs or including an ICI in a subsequent therapy doublet.²⁴¹ Nevertheless, ICIs, used alone and in combination, provide durable responses and improve OS for advanced ccRCC. ICIs have transformed the therapeutic landscape of RCC.

TARGETED MOLECULAR THERAPY

Tumor growth and angiogenesis are mediated through a number of intracellular pathways involving VEGF, HIF, and mTOR (Fig. 43.4). Treatments targeting these pathways generally inhibit the enhanced tumor vascularity and cell growth and proliferation and have revolutionized the approach to cancer therapy in many solid organs including the kidney.

VASCULAR ENDOTHELIAL GROWTH FACTOR SIGNALING PATHWAY

The first VEGF inhibitor, the anti-VEGF antibody bevacizumab, prevents angiogenesis by inhibiting binding of VEGF ligand to VEGFR, thereby inhibiting the VEGF signaling pathway. VEGF inhibitors also impede tumor cell growth directly because VEGF is present in not only endothelial cells but also many other cell types including tumor cells. VEGF signaling can also be inhibited at the receptor level using VEGFR2 tyrosine kinase inhibitors (VEGFR2 TKIs, for simplicity here: TKIs) including axitinib, cabozantinib, lenvatinib, pazopanib, sorafenib, sunitinib, and most recently, tivozanib.^{202,242} VEGFR2-targeted agents are only modestly effective as single agents and are often characterized by off-target effects and chronic toxicities, such as fatigue and rash.^{243,244} Resistance can occur from the stimulation of alternative angiogenic pathways bypassing the VEGF blockage.^{245,246} Interestingly, unlike for ICIs, there is not necessarily cross-resistance for TKIs: after progression on one, two, or three lines of TKIs, switching to yet another VEGFR2-targeted TKI can result in tumor shrinkage and stabilization.^{199,202,242} Common, dose-dependent adverse effects of VEGF signaling inhibitors, which are important for the nephrologist to know and treat, are HTN (11%–43%) and proteinuria (41%–63%) related

to the disruption of the role of VEGF in maintaining the integrity of the endothelium and slit diaphragm of the glomerular membrane.^{190,247} Kidney biopsy most frequently shows thrombotic microangiopathy, but various glomerular diseases have been reported as well.^{248,249} The drug toxicity is generally treated supportively with antihypertensive agents and protein-lowering medications²⁴⁴ (also see [Chapter 57](#)). TKIs, used as monotherapy or in combination with ICI, remain a mainstay of RCC treatment.

mTOR SIGNALING PATHWAY

Pathways involving mTOR and its upstream kinase, PI3K, also promote tumor growth. mTOR is crucial for ribosomal activity in cell metabolism, growth, and proliferation through two separate complexes (mTORC1 and mTORC2). The PI3K pathway controls growth factor expression and tumor cell proliferation. The first-generation mTOR inhibitors (everolimus, temsirolimus) suppress mTORC1 activity alone; therefore mitogenic stimulation can occur alternately through mTORC2 and PI3K. These agents can generally stabilize RCC tumors (with a response rate of 7%–26%) but rarely reduce tumor size unless mTOR pathway mutations are present.^{250,251} Newer mTOR inhibitors that additionally block the mTORC2 and PI3K pathway, have had disappointing clinical activity in RCC to date.^{252,253} Everolimus was for many years a standard therapy for refractory disease and was used as the control arm in randomized trials of newer agents including nivolumab, cabozantinib, and belzutifan. In these trials, everolimus was consistently inferior.^{199,204,254–256} Currently, with superior TKI monotherapy and combination therapies available, everolimus is rarely used as a single agent and has now been relegated to later lines of therapy.^{257,258}

HYPOXIA-INDUCIBLE FACTOR

The HIF transcription factor complex is composed of HIF2 α /HIF1 β or HIF1 α /HIF1 β heterodimers and is the effector of a cellular oxygen-sensing pathway. When pVHL function is lost, HIF is stabilized and activates transcription of genes including VEGF, erythropoietin, and glycolytic enzymes critical for metabolic regulation.^{259,260} Belzutifan is a first-in-class drug that targets HIF2 α and is the newest targeted therapy for ccRCC. This small molecule binds the hydrophobic pocket of HIF2 α , preventing heterodimerization with HIF1 β and disrupting HIF's ability to activate transcription of downstream hypoxia genes.²⁶¹ In patients with VHL syndrome, treatment with belzutifan results in significant RCC control: With a median follow-up of 21.8 months, an unprecedented 0% of patients experienced tumor progression.²⁰³ Furthermore, hemangioblastomas in the central nervous system and pancreatic neuroendocrine tumors were also well controlled. In 2021, belzutifan was approved for adult patients with VHL syndrome for the treatment of their RCC and VHL-associated tumors.

In patients with sporadic advanced ccRCC, a randomized study compared belzutifan to everolimus in patients who had previously received two lines of therapy. At the first interim analysis, the HR for progression-free survival (PFS) significantly favored belzutifan (HR = 0.75, $P < 0.001$); OS data are still not mature.²⁵⁶ Nevertheless, due to its activity in later lines of therapy, in 2023, belzutifan was approved

for patients with advanced RCC who had progressed on prior PD-1 or PD-L1 inhibitors and had also progressed on prior TKI. Adjuvant and first-line studies using belzutifan in conjunction with other agents (such as combined with ICI and/or TKI) are under way, as well as studies of additional small molecules targeting HIF2 α .

Though both VEGF signaling and HIF inhibitors target the angiogenesis signaling pathway, side effect profiles are quite different. Treatment-emergent adverse events for belzutifan are related to the fact that HIF-dependent transcription of erythropoietin is inhibited, resulting in anemia in 80% to 90% of patients, as well as fatigue in 30%, and hypoxia in 15%.^{203,262} These toxicities can be mitigated by dose reduction and use of erythropoietin. Additional side effects in approximately 20% to 30% of patients include edema, increased serum creatinine, hypoglycemia, hypocalcemia, hyponatremia, hypokalemia, lymphopenia, myalgia, and transaminitis. Side effects are generally completely reversible by holding medication and reducing dose and are much easier to tolerate compared with side effects of TKIs or potential long-term side effects of ICIs. Due to its relatively mild side effect profile and clinical efficacy, the addition of belzutifan to the therapeutic armamentarium for RCC provides a gain for patients as an alternative to TKIs and opens the door for new combinations.

THERAPY COMBINATIONS

Each of the VEGF signaling/mTOR/HIF/ICIs discussed earlier has activity as a single agent. When combined in doublets, these agents can have synergistic clinical activity. Therefore combination therapies are now routinely used in first-line therapy for advanced disease. The combination with the longest median survival follow-up (> 8 years) is the combination of dual checkpoint inhibitors, ipilimumab + nivolumab, as discussed earlier.^{154,263} Doublet combinations of TKI plus PD-1 or PD-L1 inhibitors, with reported median follow-up of 5 to 6 years, are also standard front-line therapies. The following combinations have been studied in randomized studies compared with single-agent sunitinib: axitinib/pembrolizumab,²⁰⁶ axitinib/avelumab,²⁰⁷ cabozantinib/nivolumab,^{264,265} and lenvatinib/pembrolizumab.^{209,266} OS HRs significantly favor TKI doublets compared with sunitinib, but the survival benefit does not yet seem to be as durable as for ipilimumab/nivolumab. Nevertheless, compared with ipilimumab plus nivolumab, doublets that include a TKI have a lower primary progression rate (as low as 10%), with a higher chance of tumor shrinkage and disease control. Therefore a doublet including a TKI is preferred over ipilimumab/nivolumab when patients have high tumor burden and/or require rapid tumor shrinkage. As TKI doublet studies mature, more information regarding therapy durability will emerge.

As an alternative combination regimen for patients who either cannot tolerate ICI or who have progressed on ICI,^{200,209} there is an approved combination of VEGFR2 TKI lenvatinib and mTOR inhibitor everolimus. This combination has acceptable tolerability and high efficacy. To increase efficacy even higher, triplet combinations are being studied, with careful attention to the potential for greater toxicity and efficacy. One randomized study of triple therapy has been reported to date: Ipilimumab plus nivolumab

plus cabozantinib demonstrated improved HR for PFS and higher overall response rates compared with ipilimumab plus nivolumab, as well as increased but tolerable adverse events. OS data are not yet mature.²⁶⁷ Additional studies of immune checkpoint inhibitors plus TKI plus HIF2a inhibitors are in progress (NCT06191796).²⁶⁸ The future of therapy for advanced RCC will continue to evolve as additional studies mature.

Much attention has been given to the selection of first-line therapy. However, optimal sequencing of therapies in second-line therapy and beyond have not yet been determined. In the absence of biomarkers to guide therapeutic decisions, the general approach is empiric use of any of the remaining approved agents, typically as single agents.²⁶⁹

Non-clear cell RCC (nccRCC) subtypes are generally treated with the same single agent and combination regimens as ccRCC, with smaller studies that have demonstrated efficacy.²⁷⁰ Notably, a few specific nccRCC subtypes have different preferred therapies: 1. FH-deficient RCC has excellent disease control with EGFR inhibitor erlotinib + bevacizumab²⁷¹; 2. collecting duct and renal medullary/SMARCB1-deficient subtypes are treated with standard cytotoxic chemotherapy regimens^{272,273}; and 3. ALK-rearranged RCCs respond to ALK inhibitors.²⁷⁴ Moving forward, as we better define molecular subtypes of RCC, therapies to target molecular vulnerabilities may offer improved outcomes.

PERIOPERATIVE SYSTEMIC THERAPY FOR RENAL CELL CARCINOMA

Perioperative systemic therapy can be either neoadjuvant (given before nephrectomy, to improve resectability) or adjuvant (given after nephrectomy, to decrease recurrence rate and improve survival).

NEOADJUVANT SYSTEMIC THERAPY

The goal of neoadjuvant systemic therapy is either to reduce the size of the primary tumor to preserve renal function with a nephron-sparing approach or reduce size of advanced disease to improve feasibility of eradicating all tumors. Single-agent TKIs including axitinib, pazopanib, cabozantinib, sorafenib, and sunitinib were studied in small, mostly single-arm studies of 12 to 30 patients before nephrectomy. These agents were able to decrease tumor size in up to 46% of patients (axitinib).²⁷⁵⁻²⁸² Axitinib was able to reduce the size of high-risk RCCs by an average of 1.3 cm, reduce nephrometry score by 1, and enable partial nephrectomy in 75% of patients.²⁸³ In a neoadjuvant study of 40 patients with axitinib plus avelumab, a 20% partial response rate has been reported.²⁸⁴ Prospective randomized trials specifically in the neoadjuvant setting are awaited to determine how long to treat before surgery and which regimen is most effective. So far, neoadjuvant treatment is an individualized and multidisciplinary decision, requiring close collaboration, reassessments, and discussion with medical oncologists and urologists.

ADJUVANT SYSTEMIC THERAPY

Many patients are cured with nephrectomy alone. Patients at high risk of recurrence have been selected for many adjuvant studies including single agent VEGFR2 inhibitors

(sunitinib, sorafenib, axitinib, pazopanib); mTOR inhibitor (everolimus); and ICIs (atezolizumab, pembrolizumab, nivolumab, and ipilimumab plus nivolumab).²⁸⁵⁻²⁹¹ Sunitinib was the first adjuvant treatment approved for RCC, based on an improved disease-free survival (DFS) compared with surveillance alone. However, sunitinib has not demonstrated an OS benefit in the adjuvant setting, so it is seldom used for this indication. The Keynote-564 study^{292,293} of adjuvant pembrolizumab is the first, and currently only, randomized study to demonstrate an OS benefit for patients with high-risk ccRCC (patients with pT2 grade 4, or any pT3 or pT4 or node-positive disease or fully resected metastatic disease).²⁹⁴ One year of adjuvant pembrolizumab treatment resulted in an OS HR of 0.62 (95% CI 0.44–0.87; $P = .0024$) and DFS HR of 0.72 (95% CI 0.59–0.87), both compared with placebo. Toxicities were common: The most common adverse events included fatigue and diarrhea. On the basis of efficacy and its safety profile, pembrolizumab has been approved by the FDA for adjuvant treatment of ccRCC. Notably, in contrast to the PD-1 inhibitor pembrolizumab, neither atezolizumab (PD-L1 inhibitor) nor nivolumab (PD-1 inhibitor) alone or in combination with ipilimumab have demonstrated DFS or OS benefits in their intention-to-treat cohorts. Furthermore, providers should consider that even in patients selected for high risk of recurrence, up to 40% of patients may be cured by nephrectomy alone. Thus development of biomarkers to select patients who benefit from treatment will be key to avoiding exposure of all patients to checkpoint inhibitor toxicity. With so many negative studies, newer adjuvant investigations focus on combinations, such as pembrolizumab plus belzutifan. It remains to be seen if combination therapies can affect OS in the adjuvant setting, as they have in the advanced setting.

NOVEL SYSTEMIC THERAPIES IN DEVELOPMENT

Despite advances in systemic treatment, most patients with advanced disease eventually progress on current therapies. There remains an important need for new drugs with improved efficacy and tolerability. Various new drugs and approaches are being developed for RCC, ranging from antibody conjugates to novel immunotherapies and even probiotics. In some cases, next-generation molecules are being investigated to improve upon approved first-generation molecules.²⁹⁵ Examples are TKIs that target VEGFR plus different spectra of additional kinases (XL-092 that is tuned to inhibit VEGFR, cMET, and TAM kinases); AXL inhibitors (batiraxcept); and novel HIF2a inhibitors. Described as follows are agents directed at new targets in RCC.

Drug development efforts have attempted to exploit ccRCC metabolic reprogramming by targeting critical proteins or enzymes such as glutaminase involved in the dysregulated glutamine metabolic pathway.^{296,297} Telaglenastat was a first-in-class glutaminase inhibitor with promising phase I/II data and without significant toxicity.²⁹⁸⁻³⁰¹ Unfortunately, in the randomized trials for advanced ccRCC, there was no significant PFS or OS advantage.^{302,303} As additional knowledge about other aspects of metabolic alterations in RCC arises (e.g., in fatty acid metabolism), new agents may enter development.

New immunotherapies are perhaps the largest group of drugs that are currently in development. Antibodies directed against T-cell checkpoint inhibitor proteins (LAG-3, TIM-3, TIGIT) and stimulatory T-cell checkpoint proteins (OX40 and 4-1BB) are in various phases of clinical testing³⁰⁴⁻³¹¹ (NCT03744468, NCT03628677, NCT02913313, NCT03260322, NCT03977467, NCT04457778, NCT02253992, NCT03809624, NCT02528357, NCT02705482, NCT04198766, NCT03071757). In addition, bispecific antibodies are in development for RCC, including a bispecific antibody targeting both CD47 and PD-L1 (NCT05048160). Additional strategies being explored include neoantigen vaccines (NCT03289962, NCT02950766), chimeric antigen receptor T cells [CAR-T] targeting CD70 (NCT02830724), CAR-T targeting CA9 (NCT04969354), and engineered T cells targeting HERV-E (NCT03354390). Several strategies employing agents designed to alter immune function are under active investigation; the next few years will likely see an expansion of available immunotherapies.

An alternative approach being investigated is alteration of the gut microbiome. A pilot randomized phase I study tested addition of a bifidogenic live bacterial product to standard ipilimumab/nivolumab therapy and showed improved PFS.²³⁴ Additional studies to alter the microbiome as a therapeutic strategy in RCC are under way (NCT04758507, NCT05122546).

Several agents being investigated in RCC have already been proven to have efficacy in hematologic malignancies or other solid tumors. One example includes antibody-drug conjugates, which use antibodies to deliver cytotoxic chemotherapy agents, providing “targeted chemotherapy.” Although most RCCs are not sensitive to cytotoxic chemotherapy, using an antibody to deliver highly potent chemotherapy may overcome resistance. Despite a negative phase II study in RCC (testing AGS-16C3F, an anti-ENPP3 antibody, conjugated to chemotherapy agent monomethyl auristatin F),³¹² antibody-drug conjugates directed against TIM1, CD70, or CDH-6 may reveal clinical activity (NCT02837991, NCT01015911, NCT04707248). Additional targets also include cell cycle targets and epigenetic targets. Abemaciclib and palbociclib are cyclin-dependent kinase 4/6 (CDK4/6) inhibitors in use for treatment of breast cancer.³¹³ In RCC, loss of cyclin-dependent kinase inhibitor 2A occurs in approximately a quarter of cases.^{64,314} This event results in unrestrained CDK4 activity and increased proliferation.³¹⁵ Currently, phase I/II studies are studying the activity of CDK4/6 inhibitors combined with approved RCC therapies (NCT03905889, NCT05468697, NCT05176288). Agents that impact epigenetic regulation have proven efficacy in hematologic malignancies. Preclinical data in RCC reveals that histone deacetylases (HDACs) can stabilize HIF proteins, and furthermore, HDAC inhibitors can induce HIF degradation through a VHL-independent proteasomal pathway.³¹⁶ Early-phase studies combining HDACs with angiogenesis inhibitors bevacizumab or pazopanib showed safety, tolerability, and possible activity.^{81,317} A phase III study to test addition of HDAC inhibitor abexinostat to pazopanib is currently recruiting patients. It is likely that additional targeted therapies will expand therapeutic RCC options in the future.

MOLECULAR BIOMARKERS

Prognostic and predictive molecular biomarkers have been sought for RCC through DNA, RNA, and protein profiling, but currently, none are yet sufficiently validated for clinical use.³¹⁸ Prognostic biomarkers can stratify which subpopulations have most favorable or poor outcomes. Examples of prognostic biomarkers in RCC include functional loss of BAP1, p53, SETD2, PBRM1, or overexpression of PD-L1, which are associated with decreased survival in RCC.³¹⁹⁻³²⁴ Predictive biomarkers aim to stratify which subpopulations are likely to benefit from specific therapies. Bulk RNA sequencing has revealed molecular subtypes that have “angiogenic,” “immunogenic,” “inflammatory,” or “myeloid” characteristics, which correlate with PFR and response to angiogenesis inhibitors or ICIs in their discovery cohorts.^{264,325-329} However, the signatures do not universally cross-validate when applied retrospectively to banked specimens from subsequent studies. Prospective validation will be required to demonstrate that use of a molecular signature to inform therapeutic selection affects outcome. One such study is the ongoing phase II utility of transcriptomic signatures in RCC. The shifting therapeutic landscape has proven to be a challenge for prospective validation of RCC biomarkers. With increasing numbers of therapies becoming available for patients with RCC, biomarkers to select treatments for patients who will benefit and prevent patients from receiving ineffective treatment are needed as urgently as ever.

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QUESTIONS

- The risk of renal cell carcinoma (RCC) is greatest in which of the following patients?
 - A 36-year-old man with regular alcohol use and polycystic kidney disease
 - A 46-year-old woman with coronary artery disease and congestive heart failure
 - A 70-year-old man with end-stage kidney disease on dialysis for 10 years
 - A 65-year-old female renal transplant recipient

Answer: c

Rationale: In general, older men in their sixth to seventh decade of life with diabetes mellitus, hypertension, tobacco exposure, obesity, and end-stage kidney disease, with prolonged dialysis, are at increased risk for RCC. Cystic diseases such as acquired cystic kidney disease are associated with RCC, but polycystic kidney disease is not.

- Which of the following statements regarding chronic kidney disease (CKD) and renal cell carcinoma is true?
 - Radical nephrectomy is associated with a lower risk of CKD.
 - Routine preoperative screening with urinalysis for proteinuria is recommended.
 - Kidney function assessment with creatinine is not recommended for those with small renal masses.
 - Tumor size does not determine the risk of CKD.

Answer: b

Rationale: Shared risk factors for CKD and RCC (older age, tobacco exposure, diabetes mellitus, hypertension, and obesity), in addition to a high prevalence of CKD in the small renal mass population (≤ 4 cm), predisposes them to an increased risk of new-onset CKD, CKD progression, or end-stage kidney disease (ESKD) after surgical resection. In addition, the level of estimated glomerular filtration rate (eGFR), poor nutrition (lower albumin), and surgical factors, including tumor size and acute kidney injury, have been associated with a higher incidence of posttreatment CKD. Identification of preexisting CKD (serum creatinine and eGFR calculation) and proteinuria assessment (urine albumin-to-creatinine or urine protein-to-creatinine ratios) further allows for CKD staging using the KDIGO classification criteria, as recommended by the American Urological Association guidelines. CKD risk is less with partial nephrectomy as compared with radical nephrectomy.

- Renal cell carcinoma is called “the internist’s tumor” because
 - It occurs in the internal compartment of the body.
 - Primary care internists are the most likely to treat this

disease.

- RCC has multiple and varied presentations that require the astute mind of an internist to put together and make the diagnosis.

Answer: c

Rationale: Clear cell RCC, the most common type, is a metabolic disease associated with many abnormalities in metabolic pathways. Thus because each pathway abnormality results in a discrete sign or symptom that often results from cytokine or aberrant hormone production, ccRCC often presents in a pleiotropic fashion and requires the full intellectual force of a good internist for diagnosis.

- RCC has been notoriously difficult to treat due to tumor resistance after several years. New approaches to treatment include which of the following?
 - Immune checkpoint inhibitors
 - Kinase inhibition
 - Hif2a inhibition
 - All of the above

Answer: d

Rationale: Approved therapies include immune checkpoint inhibitors targeting CTLA-4 and PD-1, Kinase inhibitors such as axitinib (targeting VEGFR), cabozantinib (targeting VEGFR and MET), and everolimus (targeting MTOR) are used alone or in combination. The newest approved therapy is the HIF inhibitor belzutifan.

- Which of the following statements concerning the genetics of ccRCC is true?
 - Most cases of ccRCC are linked to hereditary diseases.
 - Mutations of the von Hippel–Lindau (*VHL*) gene are strongly linked with sporadic ccRCC.
 - VHL* disease is predominantly associated with papillary RCC.
 - Mutations of the *VHL* gene are the only genetic abnormalities linked to ccRCC.

Answer: b

Rationale: Most cases of ccRCC are not linked to hereditary diseases but are sporadic. Mutations of the *VHL* gene are found in more than 80% of cases of sporadic ccRCC. *VHL* disease is associated with ccRCC. Other genetic abnormalities found in ccRCC include mutations in epigenetic regulators and chromatin remodeling genes and disruption of the PI3K-AKT-mTOR signaling pathways.